INVERSE TRIGONOMETRIC FUNCTIONS

Worksheet – ISC / CBSE Board Questions

Name: _____ Class: _____ Date: _____

Section A: Prove / Show That

1. Prove that $\sin^{-1}\dfrac{x}{\sqrt{1+x^2}}+\cos^{-1}\dfrac{x+1}{\sqrt{x^2+2x+2}}=\tan^{-1}(x^2+x+1)$ **[ISC]**
2. Show that $\sin^{-1}\dfrac{\sqrt{3}}{2}+2\tan^{-1}\dfrac{1}{\sqrt{3}}=\dfrac{2\pi}{3}$ **[ISC]**
3. Show that $\sin^{-1}\left(\dfrac{4}{5}\right)+\cos^{-1}\left(\dfrac{2}{\sqrt{5}}\right)=\cot^{-1}\left(\dfrac{2}{11}\right)$ **[ISC]**
4. Prove that $2\left(\tan^{-1}1+\tan^{-1}\dfrac{1}{2}+\tan^{-1}\dfrac{1}{3}\right)=\pi$ **[ISC 2004]**
5. Prove that $\cot\left(\dfrac{\pi}{4}-2\cot^{-1}3\right)=7$ **[ISC 2003]**
6. Show that $\sin^{-1}\left(\dfrac{1}{\sqrt{17}}\right)+\cos^{-1}\left(\dfrac{9}{\sqrt{85}}\right)=\tan^{-1}\left(\dfrac{1}{2}\right)$ **[ISC 2006]**
7. Prove that $2\tan^{-1}\left(\dfrac{1}{3}\right)+\cot^{-1}(4)=\tan^{-1}\dfrac{16}{13}$ **[ISC 2007]**
8. If $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=\pi$, prove that $x^2-y^2-z^2+2yz\sqrt{1-x^2}=0$ **[ISC 2009, 2024]**
9. Prove that $\sin\left[2\tan^{-1}\dfrac{3}{5}-\sin^{-1}\dfrac{7}{25}\right]=\dfrac{304}{425}$ **[ISC 2010]**
10. Prove that $2\tan^{-1}\dfrac{1}{5}+\cos^{-1}\dfrac{7}{5\sqrt{2}}+2\tan^{-1}\dfrac{1}{8}=\dfrac{\pi}{4}$ **[ISC 2011; CBSE DB 2014]**
11. Prove that $\sec^2(\tan^{-1}2)+\operatorname{cosec}^2(\cot^{-1}3)=15$ **[ISC 2012]**
12. Prove that $\cos^{-1}\dfrac{63}{65}+2\tan^{-1}\dfrac{1}{5}=\sin^{-1}\dfrac{3}{5}$ **[ISC 2012]**
13. Prove that $\tan^{-1}\dfrac{1}{4}+\tan^{-1}\dfrac{2}{9}=\dfrac{1}{2}\sin^{-1}\dfrac{4}{5}$ **[ISC 2013]**
14. If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$, prove that $x^2+y^2+z^2+2xyz=1$ **[ISC 2014]**
15. If $\sin^{-1}x+\tan^{-1}x=\dfrac{\pi}{2}$, prove that $2x^2+1=\sqrt{5}$ **[ISC 2016]**
16. Prove that $\dfrac{1}{2}\cos^{-1}\left(\dfrac{1-x}{1+x}\right)=\tan^{-1}\sqrt{x},\ x\in[0,1]$ **[ISC 2017; CBSE 2019C, 2020]**
17. If $\tan^{-1}a+\tan^{-1}b+\tan^{-1}c=\pi$, prove that $a+b+c=abc$ **[ISC 2018]**
18. If $\sec^{-1}x=\operatorname{cosec}^{-1}y$, show that $\dfrac{1}{x^2}+\dfrac{1}{y^2}=1$ **[ISC 2019]**
19. Prove that $\tan^2(\sec^{-1}2)+\cot^2(\operatorname{cosec}^{-1}3)=11$ **[ISC 2020]**
20. If $\cos^{-1}\dfrac{x}{2}+\cos^{-1}\dfrac{y}{3}=\theta$, prove that $9x^2-12xy\cos\theta+4y^2=36\sin^2\theta$ **[ISC 2020]**
21. If $\tan^{-1}\left(\dfrac{x-1}{x+1}\right)+\tan^{-1}\left(\dfrac{2x-1}{2x+1}\right)=\tan^{-1}\left(\dfrac{23}{36}\right)$, prove that $24x^2-23x-12=0$ **[ISC 2023]**
22. If $\cos^{-1}\dfrac{x}{a}+\cos^{-1}\dfrac{y}{b}=\theta$, prove that $\dfrac{x^2}{a^2}-\dfrac{2xy}{ab}\cos\theta+\dfrac{y^2}{b^2}=\sin^2\theta$ **[CBSE 2016]**
23. Show that $\tan^{-1}\left(\dfrac{1}{\sqrt{3}}\tan\dfrac{x}{2}\right)=\dfrac{1}{2}\cos^{-1}\left(\dfrac{1+2\cos x}{2+\cos x}\right)$ **[Specimen 2010]**
24. Prove that $\tan^{-1}\dfrac{1}{4}+\tan^{-1}\dfrac{2}{9}=\cos^{-1}\dfrac{2}{\sqrt{5}}$ **[ISC 1993]**
25. Prove that $2\left(\tan^{-1}\dfrac{1}{4}+\tan^{-1}\dfrac{2}{9}\right)=\tan^{-1}\dfrac{4}{3}$ **[CBSE 2008]**
26. Prove that $\tan^{-1}\dfrac{1}{5}+\tan^{-1}\dfrac{1}{7}+\tan^{-1}\dfrac{1}{3}+\tan^{-1}\dfrac{1}{8}=\dfrac{\pi}{4}$ **[ISC 1996; CBSE 2008]**
27. Prove that $4\left(\cot^{-1}3+\operatorname{cosec}^{-1}\sqrt{5}\right)=\pi$ **[WB]**
28. Prove that $\tan^{-1}\left(\frac{1}{2}\tan 2A\right)+\tan^{-1}(\cot A)+\tan^{-1}(\cot^3 A)=0$ **[ISC 2012]**
29. If $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\dfrac{\pi}{2}$, $x,y,z>0$, show that $xy+yz+zx=1$ **[CBSE 2015]**
30. Prove that $\dfrac{9\pi}{8}-\dfrac{9}{4}\sin^{-1}\dfrac{1}{3}=\dfrac{9}{4}\sin^{-1}\dfrac{2\sqrt{2}}{3}$ **[CBSE 2020; NCERT]**
31. Prove that $\sin^{-1}\dfrac{4}{5}+2\tan^{-1}\dfrac{1}{3}=\dfrac{\pi}{2}$ **[Dhanbad]**
32. Prove that $\cos^{-1}\dfrac{4}{5}=\tan^{-1}\dfrac{3}{4}$
33. Prove that $\tan^{-1}2-\tan^{-1}1=\tan^{-1}\dfrac{1}{3}$
34. Prove that $2\tan^{-1}\dfrac{1}{3}=\tan^{-1}\dfrac{3}{4}$
35. Prove that $2\tan^{-1}\dfrac{1}{3}+\tan^{-1}\dfrac{1}{7}=\dfrac{\pi}{4}$
36. Prove that $\cos^{-1}\dfrac{4}{5}+\cot^{-1}\dfrac{5}{3}=\tan^{-1}\dfrac{27}{11}$
37. Show that $\tan\left(\sin^{-1}\dfrac{\sqrt{3}}{2}-\cos^{-1}\dfrac{\sqrt{3}}{2}\right)=\dfrac{\sqrt{3}}{3}$
38. Show that $\cos\left(\tan^{-1}\dfrac{15}{8}-\sin^{-1}\dfrac{7}{25}\right)=\dfrac{297}{425}$
39. Show that $\sin\left(\sin^{-1}\dfrac{1}{2}+\cos^{-1}\dfrac{3}{5}\right)=\dfrac{3+4\sqrt{3}}{10}$
40. Show that $2\tan^{-1}\dfrac{1}{2}=\tan^{-1}\dfrac{4}{3}$

41. Show that $\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right) = \frac{4-\sqrt{7}}{3}$

42. Prove that $2\sin^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{17}{31}\right) = \frac{\pi}{4}$ **[CBSE 2016]**

43. Prove that $3\sin^{-1}x = \sin^{-1}(3x - 4x^3)$, $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ **[CBSE 2018]**

44. Prove that $\cos^{-1}\left(\frac{12}{13}\right) + \sin^{-1}\left(\frac{3}{5}\right) = \sin^{-1}\left(\frac{56}{65}\right)$ **[CBSE 2019]**

45. Prove that $\sin^{-1}\left(\frac{8}{17}\right) + \cos^{-1}\left(\frac{4}{5}\right) = \cot^{-1}\left(\frac{36}{77}\right)$ **[CBSE 2019]**

46. Prove that $\sin^{-1}\frac{4}{5} + \tan^{-1}\frac{5}{12} + \cos^{-1}\frac{63}{65} = \frac{\pi}{2}$ **[CBSE 2019]**

47. Prove that $\tan^{-1}\left(\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}}\right) = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}x$, $-\frac{1}{\sqrt{2}} \le x \le 1$ **[CBSE 2019C]**

48. Show that $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2\cos^{-1}x$, $\frac{1}{\sqrt{2}} \le x \le 1$ **[CBSE 2020]**

49. Prove that $2\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{7} = \tan^{-1}\frac{31}{17}$ **[CBSE 2020C]**

Section B: Solve for x

50. Solve $\tan^{-1}(x+2) + \tan^{-1}(2-x) = \tan^{-1}\left(\frac{2}{3}\right)$ **[ISC]**

51. Solve for x: $\tan^{-1}(x-1) + \tan^{-1}x + \tan^{-1}(x+1) = \tan^{-1}3x$ **[ISC 2008]**

52. Solve: $\cos^{-1}\left[\sin\left(\cos^{-1}x\right)\right] = \frac{\pi}{3}$ **[ISC 2010]**

53. Solve for x: $\sin(2\tan^{-1}x) = 1$ **[ISC 2011, 2019, 2020]**

54. Solve for x: $\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$, $x \neq 0$ **[ISC 2013, 2017]**

55. Solve: $\cos^{-1}\left(\sin\cos^{-1}x\right) = \frac{\pi}{6}$ **[ISC 2015]**

56. Solve for x: $\sin^{-1}\frac{5}{x} + \sin^{-1}\frac{12}{x} = \frac{\pi}{2}$, $x \neq 0$ **[ISC 2015; CBSE 2020]**

57. Solve for x, if $\tan\left(\cos^{-1}x\right) = \frac{2}{\sqrt{5}}$ **[ISC 2016]**

58. Solve: $3\tan^{-1}x + \cot^{-1}x = \pi$ **[ISC 2018]**

59. If $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$, find x **[ISC 2019; CBSE DB 2014]**

60. Solve for x: $5\tan^{-1}x + 3\cot^{-1}x = 2\pi$ **[ISC 2023]**

61. Solve for x: $\sin^{-1}\left(\frac{x}{2}\right) + \cos^{-1}x = \frac{\pi}{6}$ **[ISC 2024; Dhanbad]**

62. Solve for x: $\cos\left(\sin^{-1}x\right) = \frac{1}{9}$ **[ISC 1997]**

63. Solve: $\sin^{-1}6x + \sin^{-1}(6\sqrt{3}x) = -\frac{\pi}{2}$ **[ISC 1999]**

64. Solve: $\tan^{-1}2x + \tan^{-1}3x = \frac{\pi}{4}$ **[ISC 1995; CBSE 2008, 2015C]**

65. Solve: $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\frac{8}{31}$ **[AICBSE 2015]**

66. If $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$, find x **[CBSE DB 2015]**

67. Solve: $\tan^{-1}\left(\frac{x-2}{x-4}\right) + \tan^{-1}\left(\frac{x+2}{x+4}\right) = \frac{\pi}{4}$ **[AICBSE 2014]**

68. Solve: $2\tan^{-1}(\cos x) = \tan^{-1}(2\operatorname{cosec} x)$ **[CBSE DB 2014C]**

69. If $\sin\left[\cot^{-1}(x+1)\right] = \cos\left(\tan^{-1}x\right)$, find x **[CBSE DB 2015]**

70. Solve: $\sin^{-1}x + \sin^{-1}2x = \frac{\pi}{3}$

71. Solve: $\sin\left(\frac{1}{5}\cos^{-1}x\right) = 1$ **[MNR]**

72. Solve: $\tan^{-1}\frac{1}{2x+1} + \tan^{-1}\frac{1}{4x+1} = \tan^{-1}\frac{2}{x^2}$

73. Solve for x in terms of y:
    (i) $y = 2\sin^{-1}3x$
    (ii) $y = 3\cos^{-1}2x$
    (iii) $y = \frac{1}{2}\tan^{-1}(x+\pi)$

74. Solve for x: $\cos(\tan^{-1}x) = \sin\left(\cot^{-1}\frac{3}{4}\right)$ **[CBSE 2016]**

75. If $\tan^{-1}\left(\frac{x-3}{x-4}\right) + \tan^{-1}\left(\frac{x+3}{x+4}\right) = \frac{\pi}{4}$, find the value of x. **[CBSE 2017]**

76. Solve for x: $\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}$ **[CBSE 2020, 2020C]**

77. If $\tan^{-1}x - \cot^{-1}x = \tan^{-1}\frac{1}{\sqrt{3}}$, $x > 0$, find the value of x and hence the value of $\sec^{-1}\left(\frac{2}{x}\right)$. **[CBSE 2019]**

78. Find the value of k if $\sin^{-1}\left[k\tan\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right] = \frac{\pi}{3}$ **[CBSE 2024]**

Section C: Evaluate / Simplify / Find

79. Evaluate: $\tan\left[2\tan^{-1}\frac{1}{2} - \cot^{-1}3\right]$ **[ISC 2014; CBSE 2020]**

80. Evaluate $\cos\left(2\cos^{-1}x + \sin^{-1}x\right)$ at $x = \frac{1}{5}$ **[ISC 2020]**

81. Evaluate:
    (i) $\sin\cot^{-1}\cos\tan^{-1}x$ **[MNR 1993]**
    (ii) $\tan\left[\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}\right]$
    (iii) $\cos^{-1}x + \cos^{-1}\left[\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right]$, $\frac{1}{2} \le x \le 1$ **[JEE WB 1995]**

82. Find the value of $\tan\frac{1}{2}\left[\sin^{-1}\frac{2x}{1+x^2}+\cos^{-1}\frac{1-y^2}{1+y^2}\right]$, $|x|<1,\ y>0$ and $xy<1$ **[CBSE DB 2013]**

83. If $a_1, a_2, \ldots, a_n$ is an A.P. with common difference d, find $\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+\cdots+\tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right)\right]$, where $a_1,\ldots,a_n\in\mathbb{R}$ **[NCERT Exemplar]**

84. Simplify:
(i) $\sin(2\cos^{-1}x)$
(ii) $\cos(2\sin^{-1}x)$
(iii) $\tan(\sin^{-1}y)$
(iv) $\cos\left(\frac{1}{2}\cos^{-1}x\right)$

85. Find the value of $\tan^{-1}\sqrt{3}-\cot^{-1}(-\sqrt{3})$. **[CBSE 2018]**

86. Find the value of $\sin^{-1}\left[\sin\left(-\frac{17\pi}{8}\right)\right]$. **[CBSE 2020]**

87. Evaluate: $\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right]$ **[CBSE 2020]**

88. Find the value of $\sin\left(\cos^{-1}\frac{4}{5}+\tan^{-1}\frac{2}{3}\right)$. **[CBSE 2019]**

89. Write the value of $\cos^{-1}\left(-\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$. **[CBSE Foreign 2014]**

90. Write the principal value of $\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]$. **[CBSE AI 2014C]**

91. Find the value of $\cot\left(\frac{\pi}{2}-2\cot^{-1}\sqrt{3}\right)$. **[CBSE AI 2014C]**

92. Fill in the blank: The principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is _____. **[CBSE 2020]**

93. Fill in the blank: The range of the principal value branch of $y=\sec^{-1}x$ is _____. **[CBSE 2020]**

94. Simplify $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$, $0<x<\frac{1}{\sqrt{2}}$. **[CBSE 2021C]**

95. Express $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$, $-\frac{3\pi}{2}<x<\frac{\pi}{2}$, in the simplest form. **[CBSE 2020-21]**

96. Find the value of $\sin^{-1}\left[\sin\left(\frac{13\pi}{7}\right)\right]$. **[CBSE Sample 2022-23]**

97. Find the domain of $y=\sin^{-1}(x^2-4)$.
OR
Evaluate $\cos^{-1}\left[\cos\left(-\frac{7\pi}{3}\right)\right]$. **[CBSE 2023]**

98. Write the domain and range (principal value branch) of $f(x)=\tan^{-1}x$. **[CBSE 2023]**

99. Find the value of $\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]+\tan^{-1}1$. **[CBSE 2023]**

100. Evaluate $\sin^{-1}\left(\sin\frac{3\pi}{4}\right)+\cos^{-1}\left(\cos\frac{3\pi}{4}\right)+\tan^{-1}(1)$. **[CBSE 2023]**

101. Evaluate $3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)+2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)+\cos^{-1}(0)$. **[CBSE 2023]**

102. Find the value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]$. **[CBSE 2024]**

103. Express $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$, $-\frac{\pi}{2}<x<\frac{\pi}{2}$, in the simplest form. **[CBSE 2024]**

104. Find the principal value of $\tan^{-1}(1)+\cos^{-1}\left(-\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$. **[CBSE 2024]**

105. Evaluate: $\sec^2\left(\tan^{-1}\frac{1}{2}\right)+\operatorname{cosec}^2\left(\cot^{-1}\frac{1}{3}\right)$ **[CBSE 2024]**

106. Simplify $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$. **[CBSE 2025]**

107. Evaluate: $\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$ **[CBSE 2025]**

108. (a) Simplify: $\tan^{-1}\left(\frac{\cos 2x-\sin 2x}{\cos 2x+\sin 2x}\right)$, $0<x<\frac{\pi}{4}$.
OR
(b) Evaluate: $\tan\left(\sin^{-1}1-\cos^{-1}\left(-\frac{1}{2}\right)\right)$ **[CBSE 2026]**

Section D: Multiple Choice Questions

109. If $\cot^{-1}\frac{1}{5}+\tan^{-1}x=\frac{\pi}{2}$, then the value of x is
(a) $\frac{1}{5}$ (b) 1 (c) 0 (d) $-\frac{1}{5}$ **[ISC 2022]**

110. The expression for $\cos(\tan^{-1}x)$ is equal to
(a) $\frac{1}{\sqrt{1-x^2}}$ (b) $\frac{1}{\sqrt{1+x^2}}$
(c) $\frac{\sqrt{1-x^2}}{2}$ (d) $\sqrt{1-x^2}$ **[ISC 2022]**

111. The value of $\tan^{-1}1+\cos^{-1}\frac{1}{2}$ is
(a) $\frac{5\pi}{12}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{7\pi}{12}$ **[ISC 2022]**

112. The value of $\operatorname{cosec}\left(\sin^{-1}\left(\frac{-1}{2}\right)\right)-\sec\left(\cos^{-1}\left(\frac{-1}{2}\right)\right)$ is equal to
(a) -4 (b) 0 (c) -1 (d) 4 **[ISC 2024]**

113. If $y=\cot^{-1}x$, $x<0$, then
(a) $\frac{\pi}{2}<y\le\pi$ (b) $\frac{\pi}{2}<y<\pi$
(c) $-\frac{\pi}{2}<y<0$ (d) $-\frac{\pi}{2}\le y<0$ **[CBSE 2022]**

114. The value of $\sin^{-1}\left[\cos\left(\frac{13\pi}{5}\right)\right]$ is
(a) $-\frac{3\pi}{5}$ (b) $-\frac{\pi}{10}$ (c) $\frac{3\pi}{5}$ (d) $\frac{\pi}{10}$ **[CBSE Term I 2021-22]**

115. If $\sin^{-1}x>\cos^{-1}x$, then x should lie in the interval

(a) $\left(-1, \frac{1}{\sqrt{2}}\right)$ (b) $\left(0, \frac{1}{\sqrt{2}}\right)$
(c) $\left(\frac{1}{\sqrt{2}}, 1\right]$ (d) $\left(-\frac{1}{\sqrt{2}}, 0\right)$ **[CBSE 2022]**

116. If $\tan^{-1} x = y$, then
(a) $-1 < y < 1$ (b) $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$
(c) $-\frac{\pi}{2} < y < \frac{\pi}{2}$ (d) $y \in \left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$ **[CBSE Term I 2021-22; 2022]**

117. $\sin\left[\frac{\pi}{3} + \sin^{-1}\left(\frac{1}{2}\right)\right]$ is equal to
(a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$ **[CBSE 2023]**

118. If $f(x) = |\cos x|$, then $f\left(\frac{3\pi}{4}\right)$ is
(a) 1 (b) -1 (c) $\frac{-1}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{2}}$ **[CBSE 2023]**

119. The principal value of $\cot^{-1}(-\sqrt{3})$ is
(a) $-\frac{\pi}{6}$ (b) $\frac{\pi}{6}$ (c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$ **[CBSE 2020]**

120. $\tan^{-1} 3 + \tan^{-1} \lambda = \tan^{-1}\left(\frac{3+\lambda}{1-3\lambda}\right)$ is valid for what values of λ?
(a) $\lambda \in \left(-\frac{1}{3}, \frac{1}{3}\right)$ (b) $\lambda > \frac{1}{3}$
(c) $\lambda < \frac{1}{3}$ (d) all real λ **[CBSE 2020]**

121. The principal value of $\tan^{-1}\left(\tan\frac{3\pi}{5}\right)$ is
(a) $\frac{2\pi}{5}$ (b) $-\frac{2\pi}{5}$ (c) $\frac{3\pi}{5}$ (d) $-\frac{3\pi}{5}$ **[CBSE 2020]**

122. $\sin(\tan^{-1} x)$, where $|x| < 1$, is equal to
(a) $\frac{x}{\sqrt{1-x^2}}$ (b) $\frac{1}{\sqrt{1-x^2}}$
(c) $\frac{1}{\sqrt{1+x^2}}$ (d) $\frac{x}{\sqrt{1+x^2}}$ **[CBSE Term I 2021-22]**

123. The simplest form of $\tan^{-1}\left(\frac{\sqrt{1+\cos x}+\sqrt{1-\cos x}}{\sqrt{1+\cos x}-\sqrt{1-\cos x}}\right)$, $\pi < x < \frac{3\pi}{2}$, is
(a) $\frac{\pi}{4} - \frac{x}{2}$ (b) $\frac{3\pi}{2} - \frac{x}{2}$ (c) $-\frac{x}{2}$ (d) $\pi - \frac{x}{2}$ **[CBSE Term I 2021-22]**

124. If $2\cos^{-1} x = y$, then
(a) $0 \le y \le \pi$ (b) $-\pi \le y \le \pi$
(c) $0 \le y \le 2\pi$ (d) $-\pi \le y \le 0$ **[CBSE 2026]**

125. The principal value of $\sin^{-1}\left(\sin\left(-\frac{10\pi}{3}\right)\right)$ is
(a) $-\frac{2\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{\pi}{3}$ (d) $-\frac{\pi}{3}$ **[CBSE 2025]**

126. $\sec^{-1}(-\sqrt{2}) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ is equal to
(a) $\frac{11\pi}{12}$ (b) $\frac{5\pi}{12}$ (c) $\frac{7\pi}{12}$ (d) $-\frac{5\pi}{12}$ **[CBSE 2025]**

127. If $y = \sin^{-1} x$, $-1 \le x \le 0$, then the range of y is
(a) $\left(-\frac{\pi}{2}, 0\right]$ (b) $\left[-\frac{\pi}{2}, 0\right]$
(c) $\left(-\frac{\pi}{2}, 0\right)$ (d) $\left[-\frac{\pi}{2}, 0\right)$ **[CBSE 2025]**

128. The principal value of $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ is
(a) $\frac{\pi}{3}$ (b) $-\frac{2\pi}{3}$ (c) $-\frac{\pi}{3}$ (d) $\frac{2\pi}{3}$ **[CBSE 2025]**

Section E: Assertion–Reason Questions

Assertion–Reason options: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

129. **Assertion (A):** All trigonometric functions have their inverses over their respective domains.
Reason (R): The inverse of $\tan^{-1} x$ exists for some $x \in \mathbb{R}$. **[CBSE 2023]**

130. **Assertion (A):** The maximum value of $(\cos^{-1} x)^2$ is π^2.
Reason (R): The range of the principal value branch of $\cos^{-1} x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. **[CBSE 2023]**

131. **Assertion (A):** The range of $\left[\sin^{-1} x + 2\cos^{-1} x\right]$ is $[0, \pi]$.
Reason (R): The principal value branch of $\sin^{-1} x$ has range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. **[CBSE 2023]**

132. **Assertion (A):** The range of $f(x) = 2\sin^{-1} x + \frac{3\pi}{2}$, where $x \in [-1, 1]$, is $\left[\frac{\pi}{2}, \frac{5\pi}{2}\right]$.
Reason (R): The range of the principal value branch of $\sin^{-1} x$ is $[0, \pi]$. **[CBSE 2023]**

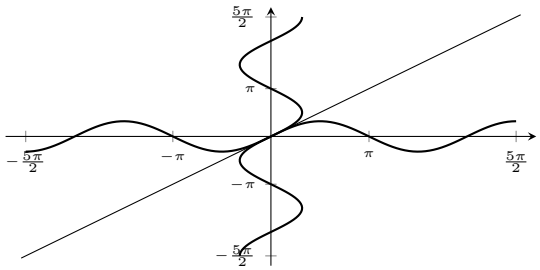
133. **Assertion (A):** The domain of the function $\sec^{-1} 2x$ is $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$.
Reason (R): $\sec^{-1}(-2) = -\frac{\pi}{4}$. **[CBSE Sample 2022-23]**

134. **Assertion (A):** The domain of $y = \cos^{-1}(x)$ is $[-1, 1]$.
Reason (R): The range of the principal value branch of $y = \cos^{-1}(x)$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$. **[CBSE 2024]**

135. **Assertion (A):** The set of values of $\sec^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is a null set.
Reason (R): $\sec^{-1} x$ is defined for $x \in \mathbb{R} - (-1, 1)$. **[CBSE 2025]**

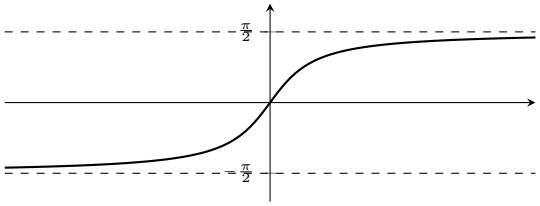
Section F: Graph-Based Questions

136. The following graph is a combination of:



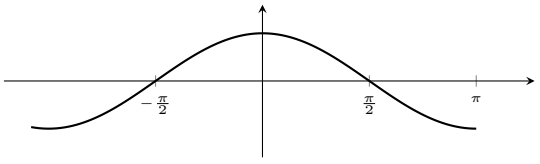
(a) $y = \sin^{-1} x$ and $y = \cos^{-1} x$ (b) $y = \cos^{-1} x$ and $y = \cos x$
(c) $y = \sin^{-1} x$ and $y = \sin x$ (d) $y = \cos^{-1} x$ and $y = \sin x$ **[CBSE 2025]**

137. The given graph illustrates:

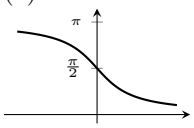


(a) $y = \sec^{-1} x$ (b) $y = \cot^{-1} x$ (c) $y = \tan^{-1} x$
(d) $y = \operatorname{cosec}^{-1} x$ **[CBSE 2025]**

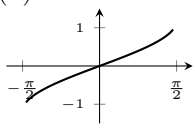
138. The graph of a trigonometric function is as shown. Which of the following will represent the graph of its inverse?



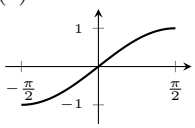
(a)



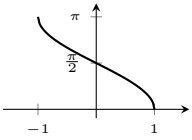
(b)



(c)



(d)



[CBSE 2025]

139. Draw the graph of $f(x) = \sin^{-1} x$, $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$. Also write the range of $f(x)$. **[CBSE 2023]**

Section G: Case Study

If a function $f : X \to Y$ defined as $f(x) = y$ is one-one and onto, then we can define a unique function $g : Y \to X$ such that $g(y) = x$, where $x \in X$ and $y = f(x)$, $y \in Y$. Function g is called the inverse of function f.

The domain of the sine function is $\mathbb{R}$ and $\sin : \mathbb{R} \to \mathbb{R}$ is neither one-one nor onto. Let the sine function be defined from set A to $[-1, 1]$ such that the inverse of the sine function exists, i.e. $\sin^{-1} x$ is defined from $[-1, 1]$ to A.

On the basis of the above information, answer the following: **[CBSE 2024]**

(i) If A is an interval other than the principal value branch, give an example of one such interval. [1]

(ii) If $\sin^{-1} x$ is defined from $[-1, 1]$ to its principal value branch, find the value of $\sin^{-1}\left(-\frac{1}{2}\right) - \sin^{-1}(1)$. [1]

(iii) (a) Draw the graph of $\sin^{-1} x$ from $[-1, 1]$ to its principal value branch.

OR

(b) Find the domain and range of $f(x) = 2\sin^{-1}(1 - x)$. [2]

Section H: Competency-Based Questions

140. If $a + \frac{\pi}{2} < 2\tan^{-1} x + 3\cot^{-1} x < b$, then a and b are respectively:
(a) $\frac{\pi}{2}$ and 2π (b) $\frac{\pi}{2}$ and $-\frac{\pi}{2}$
(c) 0 and π (d) 0 and 2π **[Competency Based – Analysis]**

141. Which one of the following is *true*?
(a) $\sin(\cos^{-1} x) = \cos(\sin^{-1} x)$
(b) $\sec(\tan^{-1} x) = \tan(\sec^{-1} x)$
(c) $\cos(\tan^{-1} x) = \tan(\cos^{-1} x)$
(d) $\tan(\sin^{-1} x) = \sin(\tan^{-1} x)$ **[Competency Based – Application]**

142. If the domain of the function $f(x) = \sqrt{\cos^{-1}(3x) + \frac{\pi}{4}}$ is $[a, b]$, then find the value of $a + b$. **[Competency Based – Analysis & Evaluate]**